

BIERE, Switzerland

WMO: 067040

Lat: 46.5250N

Lon: 6.3425E

Elev: 684

StdP: 93.37

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.8	-6.5	-12.8	1.3	-6.6	-10.0	1.7	-4.7	10.8	1.0	10.0	-2.3	4.0	40	0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	29.2	19.6	27.3	18.9	25.6	18.0	20.5	27.2	19.6	26.0	18.7	24.5	2.9	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	14.2	23.6	17.3	13.4	22.4	16.6	12.8	21.5	62.1	27.5	58.9	25.9	56.0	24.6	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.9	7.6	6.5	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-10.2	32.3	2.0	2.1	-11.7	33.8	-12.9	35.1	-14.0	36.3	-15.4	37.9		
			-11.0	22.4	2.2	1.3	-12.6	23.3	-13.9	24.0	-15.2	24.8	-16.8	25.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	1.2	1.2	5.9	9.6	12.2	16.8	18.9	18.1	14.6	10.3	5.1	1.8		
	DBStd	7.25	3.60	4.34	3.80	3.66	3.53	3.52	3.30	3.23	2.95	3.45	3.63	3.52		
	HDD10.0	1172	273	248	133	51	17	1	0	0	2	40	151	256		
	HDD18.3	3272	532	481	385	261	191	70	33	44	117	249	397	514		
	CDD10.0	1056	0	1	7	39	86	204	275	252	139	49	4	0		
	CDD18.3	116	0	0	0	0	1	23	49	38	4	0	0	0		
	CDH23.3	993	0	0	0	1	16	208	434	301	33	0	0	0		
	CDH26.7	231	0	0	0	0	0	46	121	64	1	0	0	0		
(14) Wind	WSAvg	2.6	2.8	3.0	3.1	2.8	2.7	2.4	2.3	2.1	2.3	2.3	2.5	2.7		
(15) Precipitation	PrecAvg	1272	104	99	87	92	107	107	108	107	104	115	122	122		
	PrecMax	1542	270	311	353	160	187	216	219	246	252	220	320	289		
	PrecMin	945	20	14	21	21	34	43	49	20	38	25	14	3		
	PrecStd	171	62	66	70	46	43	46	41	47	58	55	71	65		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.5	13.9	17.7	22.8	25.2	30.1	32.6	30.1	26.1	20.9	15.5	11.8		
		MCWB	7.6	7.7	9.6	13.0	15.9	21.3	19.2	19.6	17.1	14.5	11.1	7.8		
	2%	DB	9.3	10.7	16.0	20.4	23.1	28.0	29.7	28.4	24.2	19.2	13.8	9.7		
		MCWB	7.1	6.3	9.1	11.9	15.3	19.7	19.2	19.6	17.1	13.7	10.5	7.5		
	5%	DB	8.0	8.8	14.2	18.6	21.3	25.7	27.6	26.7	22.2	17.3	12.2	8.5		
		MCWB	6.4	5.3	8.3	11.2	14.1	18.4	18.5	19.1	16.1	13.5	9.9	6.8		
	10%	DB	6.4	7.0	12.0	16.4	19.2	23.7	25.6	24.7	20.4	15.5	10.7	7.3		
		MCWB	5.2	4.7	7.3	10.3	13.2	17.3	17.8	18.2	15.5	12.7	8.9	5.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	8.5	10.7	14.2	17.4	22.7	20.9	21.2	18.7	16.0	12.3	9.2		
		MCDB	10.1	11.1	15.8	21.1	23.2	29.3	28.6	27.6	22.6	18.6	13.8	10.3		
	2%	WB	7.6	7.1	9.7	12.7	16.0	20.4	20.0	20.3	17.7	14.8	11.2	8.1		
		MCDB	8.8	10.3	14.2	19.1	21.9	26.4	27.7	26.8	22.6	17.3	12.9	9.3		
	5%	WB	6.4	5.9	8.8	11.6	14.9	19.1	19.2	19.5	16.9	14.0	10.2	7.2		
		MCDB	7.8	8.3	13.2	17.3	19.4	24.8	26.2	25.5	21.4	16.5	12.0	8.6		
	10%	WB	5.3	4.7	7.8	10.6	13.7	17.8	18.5	18.6	15.9	13.1	9.1	6.0		
		MCDB	6.5	6.9	11.6	15.7	18.3	22.9	24.6	23.9	19.7	15.2	10.3	7.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	6.7	8.3	9.7	9.4	10.0	10.5	10.5	9.5	7.9	6.0	6.1		
		MCDBR	6.7	10.3	11.9	13.0	12.7	12.9	13.9	12.9	12.1	10.4	8.1	7.5		
	5% WB	MCWBR	4.9	6.7	6.6	6.7	6.5	6.7	6.0	5.9	6.3	6.3	5.7	5.6		
		MCWBR	6.2	9.0	10.7	12.1	11.6	12.4	12.6	12.3	10.8	9.3	7.7	6.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.283	0.306	0.341	0.376	0.385	0.386	0.384	0.376	0.357	0.340	0.308	0.282		
		taud	2.457	2.410	2.381	2.309	2.326	2.359	2.356	2.389	2.426	2.466	2.501	2.475		
	Ebn at Noon	Ebn at Noon	824	869	884	879	881	879	876	869	856	818	785	784		
		Edh at Noon	66	85	102	121	123	120	119	110	97	81	64	59		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.34	2.21	3.35	4.39	5.02	5.96	5.84	5.08	3.98	2.56	1.43	1.09		
	RadStd	RadStd	0.15	0.32	0.47	0.62	0.45	0.56	0.50	0.45	0.34	0.26	0.18	0.17		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (90 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	+0.37	+0.27	N/A	N/A	N/A	N/A

Nomenclature: See separate page